

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“JnanaSangama”, Belgaum- 590014, Karnataka.



LAB RECORD

on

Big Data Analytics (23CS6PCBDA)

Submitted by

S Nagashree (1BM23CS280)

in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING



B.M.S. COLLEGE OF ENGINEERING

(Autonomous Institution under VTU)

BENGALURU - 560019

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B.M.S. College of Engineering

Bull Temple Road, Bangalore 560019
(Affiliated to Visvesvaraya Technological University, Belgaum)

Department of Computer Science and Engineering



CERTIFICATE

This is to certify that the Lab work entitled “Big Data Analytics” carried out by **S Nagashree (1BM23CS280)**, who is Bonafide student of **B.M.S. College of Engineering**. It is in partial fulfilment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2026. The Lab report has been approved as it satisfies the academic requirements in respect of a Big Data Analytics (23CS6PCBDA) work prescribed for the said degree.

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GitHub Link: https://github.com/S-Nagashree/BDA_LAB_1BM23CS280.git

Course Outcomes (COs):

CO1	Apply the concept of NoSQL, Hadoop or Spark for a given task
CO2	Analyse big data analytics mechanisms that can be applied to obtain solution for a given problem.
CO3	Design and implement solutions using data analytics mechanisms for a given problem.

LABORATORY PROGRAM – 1

MongoDB- CRUD Operations Demonstration (Practice and Self Study)

COMMAND WITH OUTPUT - USING ATLAS

```
>_MONGOSH
{
  _id: ObjectId('6981be87fc16c71af8f2792b'),
  accid: 'ACC006',
  postid: 'POST016',
  likes: 470,
  comment: {
    count: 35,
    latest: 'Excellent'
  },
  shares: 70,
  pagevisit: 1200,
  engagement: 1740
}
{
  _id: ObjectId('6981be87fc16c71af8f27925'),
  accid: 'ACC007',
  postid: 'POST010',
  likes: 400,
  comment: {
    count: 25,
    latest: 'Fantastic'
  },
  shares: 60,
  pagevisit: 1000,
  engagement: 1460
}
> db.Post.aggregate([
  { $setWindowFields: {
    sortBy: { likes: -1 },
    output: { like_rank: { $rank: {} } }
  }}
]);
< {
  _id: ObjectId('6981be87fc16c71af8f27921'),
  accid: 'ACC005',
  postid: 'POST006',
  shares: 1,
  pagevisit: 50,
  like_rank: 20
}
> db.Post.aggregate([
  { $group: { _id: null, min_page_visits: { $min: "$pagevisit" } } }
]);
< {
  _id: null,
  min_page_visits: 50
}
> db.Post.aggregate([
  { $group: { _id: null, max_shares: { $max: "$shares" } } }
]);
< {
  _id: null,
  max_shares: 90
}
test>
```

```

> // Create "Post" collection and insert documents
db.Post.insertMany([
  { accid: "ACC001", postId: "POST001", likes: 120, comment: { count: 5, latest: "Nice post" }, shares: 15, pagevisit: 300 },
  { accid: "ACC002", postId: "POST002", likes: 45, comment: { count: 2, latest: "Cool" }, shares: 5, pagevisit: 120 },
  { accid: "ACC003", postId: "POST003", likes: 300, comment: { count: 20, latest: "Amazing!" }, shares: 40, pagevisit: 900 },
  { accid: "ACC004", postId: "POST004", likes: 75, comment: { count: 3, latest: "Good info" }, shares: 10, pagevisit: 200 },
  { accid: "ACC005", postId: "POST005", likes: 10, comment: { count: 1, latest: "Okay" }, shares: 1, pagevisit: 50 },
  { accid: "ACC006", postId: "POST006", likes: 500, comment: { count: 45, latest: "Loved it" }, shares: 90, pagevisit: 1500 },
  { accid: "ACC007", postId: "POST007", likes: 98, comment: { count: 6, latest: "Helpful" }, shares: 12, pagevisit: 260 },
  { accid: "ACC008", postId: "POST008", likes: 210, comment: { count: 18, latest: "Great" }, shares: 30, pagevisit: 700 },
  { accid: "ACC009", postId: "POST009", likes: 33, comment: { count: 2, latest: "Nice" }, shares: 4, pagevisit: 110 },
  { accid: "ACC010", postId: "POST010", likes: 400, comment: { count: 25, latest: "Fantastic" }, shares: 60, pagevisit: 1000 },
  { accid: "ACC011", postId: "POST011", likes: 67, comment: { count: 4, latest: "Interesting" }, shares: 8, pagevisit: 180 },
  { accid: "ACC012", postId: "POST012", likes: 150, comment: { count: 10, latest: "Well done" }, shares: 20, pagevisit: 500 },
  { accid: "ACC013", postId: "POST013", likes: 22, comment: { count: 1, latest: "Okay" }, shares: 2, pagevisit: 90 },
  { accid: "ACC014", postId: "POST014", likes: 305, comment: { count: 22, latest: "Super" }, shares: 50, pagevisit: 850 },
  { accid: "ACC015", postId: "POST015", likes: 88, comment: { count: 5, latest: "Nice work" }, shares: 11, pagevisit: 230 },
  { accid: "ACC016", postId: "POST016", likes: 470, comment: { count: 35, latest: "Excellent" }, shares: 70, pagevisit: 1200 },
  { accid: "ACC017", postId: "POST017", likes: 15, comment: { count: 1, latest: "Hmm" }, shares: 1, pagevisit: 60 },
  { accid: "ACC018", postId: "POST018", likes: 260, comment: { count: 19, latest: "Very good" }, shares: 33, pagevisit: 780 },
  { accid: "ACC019", postId: "POST019", likes: 190, comment: { count: 14, latest: "Loved this" }, shares: 25, pagevisit: 620 },
  { accid: "ACC020", postId: "POST020", likes: 55, comment: { count: 3, latest: "Good" }, shares: 6, pagevisit: 170 }
]);
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6981be87fc16c71af8f2791c'),
    '1': ObjectId('6981be87fc16c71af8f2791d'),
    '2': ObjectId('6981be87fc16c71af8f2791e'),
    '3': ObjectId('6981be87fc16c71af8f2791f'),
    '4': ObjectId('6981be87fc16c71af8f27920'),
    '5': ObjectId('6981be87fc16c71af8f27921'),
    '6': ObjectId('6981be87fc16c71af8f27922'),
    '7': ObjectId('6981be87fc16c71af8f27923'),
    '8': ObjectId('6981be87fc16c71af8f27924'),
    '9': ObjectId('6981be87fc16c71af8f27925'),
    '10': ObjectId('6981be87fc16c71af8f27926'),
  }
}

```

MongoDB- CRUD Operations Demonstration (Practice and Self Study)

COMMAND WITH OUTPUT - USING UBUNTU TERMINAL

The image displays two screenshots of MongoDB tools. The top screenshot shows the MongoDB Shell interface with several SQL queries and their corresponding results in a table format. The bottom screenshot shows the MongoDB Compass interface with the same queries and a partial view of the results.

Top Screenshot (MongoDB Shell):

```
1 • select * from Post;
2
3 • SELECT accid, SUM(likes) AS total_likes
4   FROM Post
5   GROUP BY accid
6   HAVING SUM(likes) > (
7     SELECT AVG(likes) FROM Post
8   );
9
10 • SELECT postid,
11      (likes + shares + pagevisit) AS engagement
12   FROM Post
13   ORDER BY engagement DESC
14   LIMIT 3;
15
16 • SELECT p.postid, p.accid, p.likes
17   FROM Post p
18   WHERE p.likes > (
19     SELECT AVG(likes)
20     FROM Post
```

Results Table (from top screenshot):

postid	accid	likes	like_rank
POST006	ACC005	560	1
POST016	ACC006	470	2
POST010	ACC007	400	3
POST014	ACC004	305	4
POST003	ACC003	300	5
POST018	ACC007	260	6
POST008	ACC006	210	7
POST019	ACC008	190	8
POST012	ACC008	190	9
POST001	ACC001	120	10
POST007	ACC002	98	11
POST015	ACC005	88	12
POST004	ACC001	75	13
POST011	ACC001	67	14

Bottom Screenshot (MongoDB Compass):

The bottom screenshot shows the MongoDB Compass interface with the same queries. The results grid is partially visible, showing a table with columns `max_shares` and `90`.

1BF24CS266*

Limit to 1000 rows

```

20     FROM Post
21     WHERE accid = p.accid
22 );
23
24 • SELECT postid, accid, likes,
25       RANK() OVER (ORDER BY likes DESC) AS like_rank
26 FROM Post;
27
28
29
30 • SELECT postid, accid, likes,
31       RANK() OVER (ORDER BY likes DESC) AS like_rank
32 FROM Post;
33
34 • SELECT MIN(pagevisit) AS min_page_visits
35 FROM Post;
36
37 • SELECT MAX(shares) AS max_shares
38 FROM Post;

```

Result Grid

	accid	total_likes
▶	ACC001	262
	ACC003	333
	ACC004	315
	ACC005	648
	ACC006	680
	ACC007	660
	ACC008	340

LABORATORY PROGRAM – 2

Perform the following DB operations using Cassandra

Questions:

- Create a keyspace by name Employee
- Create a column family by name
 - Employee-Info with attributes
 - Emp_Id Primary Key, Emp_Name,
 - Designation, Date_of_Joining,
 - Salary, Dept_Name
- Insert the values into the table in batch
- Update Employee name and Department of Emp-Id 121
- Sort the details of Employee records based on salary
- Alter the schema of the table Employee_Info to add a column Projects which stores a set of Projects done by the corresponding Employee.
- Update the altered table to add project names.
- Create a TTL of 15 seconds to display the values of Employees.

COMMAND WITH OUTPUT

```
cqlsh:emp1> CREATE TABLE Employee_Info (
...     Emp_Id int PRIMARY KEY,
...     Emp_Name text,
...     Designation text,
...     Date_of_Joining date,
...     Salary decimal,
...     Dept_Name text
... );
cqlsh:emp1> BEGIN BATCH
...     INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
...     VALUES (121, 'Alice Smith', 'Lead Engineer', '2022-01-15', 95000, 'R&D');
...
...     INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
...     VALUES (122, 'Bob Jones', 'Senior Analyst', '2021-03-22', 82000, 'Finance');
... APPLY BATCH;
cqlsh:emp1> UPDATE Employee_Info
... SET Emp_Name = 'Alice Johnson', Dept_Name = 'Cloud Ops'
... WHERE Emp_Id = 121;
cqlsh:emp1> SELECT * FROM Employee_Info ALLOW FILTERING;
```

emp_id	date_of_joining	dept_name	designation	emp_name	salary
122	2021-03-22	Finance	Senior Analyst	Bob Jones	82000
121	2022-01-15	Cloud Ops	Lead Engineer	Alice Johnson	95000

(2 rows)

```
cqlsh:emp1> ALTER TABLE Employee_Info ADD Projects set<text>;
cqlsh:emp1> UPDATE Employee_Info
... SET Projects = Projects + {'Project Apollo', 'Project Zeus'}
... WHERE Emp_Id = 121;
cqlsh:emp1> select * from Employee_Info;
```

emp_id	date_of_joining	dept_name	designation	emp_name	projects	salary
122	2021-03-22	Finance	Senior Analyst	Bob Jones	null	82000
121	2022-01-15	Cloud Ops	Lead Engineer	Alice Johnson	{'Project Apollo', 'Project Zeus'}	95000

(2 rows)

```
cqlsh:emp1> INSERT INTO Employee_Info (Emp_Id, Emp_Name, Salary)
... VALUES (123, 'Temporary User', 50000)
... USING TTL 15;
cqlsh:emp1> select * from Employee_Info;
```

emp_id	date_of_joining	dept_name	designation	emp_name	projects	salary
123	null	null	null	Temporary User	null	50000
122	2021-03-22	Finance	Senior Analyst	Bob Jones	null	82000
121	2022-01-15	Cloud Ops	Lead Engineer	Alice Johnson	{'Project Apollo', 'Project Zeus'}	95000

(3 rows)

```

(2 rows)
cqlsh:emp1> ALTER TABLE Employee_Info ADD Projects set<text>;
cqlsh:emp1> UPDATE Employee_Info
... SET Projects = Projects + {'Project Apollo', 'Project Zeus'}
... WHERE Emp_Id = 121;
cqlsh:emp1> select * from Employee_Info;

emp_id | date_of_joining | dept_name | designation | emp_name | projects | salary
-----+-----+-----+-----+-----+-----+-----
122 | 2021-03-22 | Finance | Senior Analyst | Bob Jones | null | 82000
121 | 2022-01-15 | Cloud Ops | Lead Engineer | Alice Johnson | {'Project Apollo', 'Project Zeus'} | 95000

(2 rows)
cqlsh:emp1> INSERT INTO Employee_Info (Emp_Id, Emp_Name, Salary)
... VALUES (123, 'Temporary User', 50000)
... USING TTL 15;
cqlsh:emp1> select * from Employee_Info;

emp_id | date_of_joining | dept_name | designation | emp_name | projects | salary
-----+-----+-----+-----+-----+-----+-----
123 | null | null | null | Temporary User | null | 50000
122 | 2021-03-22 | Finance | Senior Analyst | Bob Jones | null | 82000
121 | 2022-01-15 | Cloud Ops | Lead Engineer | Alice Johnson | {'Project Apollo', 'Project Zeus'} | 95000

(3 rows)
cqlsh:emp1> select * from Employee_Info;

emp_id | date_of_joining | dept_name | designation | emp_name | projects | salary
-----+-----+-----+-----+-----+-----+-----
122 | 2021-03-22 | Finance | Senior Analyst | Bob Jones | null | 82000
121 | 2022-01-15 | Cloud Ops | Lead Engineer | Alice Johnson | {'Project Apollo', 'Project Zeus'} | 95000

(2 rows)
cqlsh:emp1> UPDATE Employee_Info
... USING TTL 15
... SET Salary = 0
... WHERE Emp_Id = 121;
cqlsh:emp1> select * from Employee_Info;

emp_id | date_of_joining | dept_name | designation | emp_name | projects | salary
-----+-----+-----+-----+-----+-----+-----
122 | 2021-03-22 | Finance | Senior Analyst | Bob Jones | null | 82000
121 | 2022-01-15 | Cloud Ops | Lead Engineer | Alice Johnson | {'Project Apollo', 'Project Zeus'} | 0

(2 rows)
cqlsh:emp1> select * from Employee_Info;

emp_id | date_of_joining | dept_name | designation | emp_name | projects | salary
-----+-----+-----+-----+-----+-----+-----
122 | 2021-03-22 | Finance | Senior Analyst | Bob Jones | null | 82000
121 | 2022-01-15 | Cloud Ops | Lead Engineer | Alice Johnson | {'Project Apollo', 'Project Zeus'} | null

(2 rows)
cqlsh:emp1>

```

CODE:

CQLSH

CREATE KEYSPACE Employee

WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};

USE Employee;

cqlsh:emp1> CREATE TABLE Employee_Info (

... Emp_Id int PRIMARY KEY,

... Emp_Name text,

... Designation text,

... Date_of_Joining date,

... Salary decimal,

... Dept_Name text

...);

cqlsh:emp1> BEGIN BATCH

... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)

... VALUES (121, 'Alice Smith', 'Lead Engineer', '2022-01-15', 95000, 'R&D');

...

... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)

```

... VALUES (122, 'Bob Jones', 'Senior Analyst', '2021-03-22', 82000, 'Finance');
... APPLY BATCH;
cqlsh:emp1> UPDATE Employee_Info
... SET Emp_Name = 'Alice Johnson', Dept_Name = 'Cloud Ops'
... WHERE Emp_Id = 121;
cqlsh:emp1> SELECT * FROM Employee_Info ALLOW FILTERING;

```

emp_id	date_of_joining	dept_name	designation	emp_name	salary
122	2021-03-22	Finance	Senior Analyst	Bob Jones	82000
121	2022-01-15	Cloud Ops	Lead Engineer	Alice Johnson	95000

```

(2 rows)
cqlsh:emp1> ALTER TABLE Employee_Info ADD Projects set<text>;
cqlsh:emp1> UPDATE Employee_Info
... SET Projects = Projects + {'Project Apollo', 'Project Zeus'}
... WHERE Emp_Id = 121;
cqlsh:emp1> select * from Employee_Info;

```

emp_id	date_of_joining	dept_name	designation	emp_name	projects	salary
122	2021-03-22	Finance	Senior Analyst	Bob Jones	null	82000
121	2022-01-15	Cloud Ops	Lead Engineer	Alice Johnson	{'Project Apollo', 'Project Zeus'}	95000

```

(2 rows)
cqlsh:emp1> INSERT INTO Employee_Info (Emp_Id, Emp_Name, Salary)
... VALUES (123, 'Temporary User', 50000)
... USING TTL 15;
cqlsh:emp1> select * from Employee_Info;

```

emp_id	date_of_joining	dept_name	designation	emp_name	projects	salary
123	null	null	null	Temporary User	null	50000
122	2021-03-22	Finance	Senior Analyst	Bob Jones	null	82000
121	2022-01-15	Cloud Ops	Lead Engineer	Alice Johnson	{'Project Apollo', 'Project Zeus'}	95000

```

(3 rows)
cqlsh:emp1> select * from Employee_Info;

```

emp_id	date_of_joining	dept_name	designation	emp_name	projects	salary
122	2021-03-22	Finance	Senior Analyst	Bob Jones	null	82000
121	2022-01-15	Cloud Ops	Lead Engineer	Alice Johnson	{'Project Apollo', 'Project Zeus'}	95000

```

(2 rows)
cqlsh:emp1> UPDATE Employee_Info
... USING TTL 15
... SET Salary = 0
... WHERE Emp_Id = 121;
cqlsh:emp1> select * from Employee_Info;

```

emp_id	date_of_joining	dept_name	designation	emp_name	projects	salary
122	2021-03-22	Finance	Senior Analyst	Bob Jones	null	82000
121	2022-01-15	Cloud Ops	Lead Engineer	Alice Johnson	{'Project Apollo', 'Project Zeus'}	0

```

(2 rows)
cqlsh:emp1> select * from Employee_Info;

```

emp_id	date_of_joining	dept_name	designation	emp_name	projects	salary
--------	-----------------	-----------	-------------	----------	----------	--------

122	2021-03-22	Finance	Senior Analyst	Bob Jones	null	82000
121	2022-01-15	Cloud Ops	Lead Engineer	Alice Johnson	{'Project Apollo', 'Project Zeus'}	null

(2 rows)

LABORATORY PROGRAM – 3

Perform the following DB operations using Cassandra

Questions:

- a) Create a keyspace by name Library
- b) Create a column family by name Library-Info with attributes
 - Stud_Id Primary Key,
 - Counter_value of type Counter,
 - Stud_Name, Book-Name, Book-Id,
 - Date_of_issue
- c) Insert the values into the table in batch
- d) Display the details of the table created and increase the value of the counter
- e) Write a query to show that a student with id 112 has taken a book “BDA” 2 times.
- f) Export the created column to a csv file
- g) Import a given csv dataset from local file system into Cassandra column family

COMMAND WITH OUTPUT

```
cqlsh> CREATE KEYSPACE Library
... WITH replication = {
...   'class': 'SimpleStrategy',
...   'replication_factor': 1
... };
cqlsh> USE Library;
cqlsh:library> CREATE TABLE Library_Info (
...   Stud_Id INT,
...   Book_Id INT,
...   Stud_Name TEXT,
...   Book_Name TEXT,
...   Date_of_issue DATE,
...   PRIMARY KEY (Stud_Id, Book_Id)
... );
cqlsh:library> CREATE TABLE Library_Counter (
...   Stud_Id INT,
...   Book_Name TEXT,
...   Counter_value COUNTER,
...   PRIMARY KEY (Stud_Id, Book_Name)
... );
cqlsh:library> BEGIN BATCH
...
... INSERT INTO Library_Info (Stud_Id, Book_Id, Stud_Name, Book_Name, Date_of_issue)
... VALUES (112, 101, 'Rahul', 'BDA', '2026-02-24');
...
... INSERT INTO Library_Info (Stud_Id, Book_Id, Stud_Name, Book_Name, Date_of_issue)
... VALUES (113, 102, 'Anita', 'AI', '2026-02-24');
...
... APPLY BATCH;
cqlsh:library> UPDATE Library_Counter
... SET Counter_value = Counter_value + 1
```

```
... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library> SELECT * FROM Library_Info;
```

stud_id	book_id	book_name	date_of_issue	stud_name
113	102	AI	2026-02-24	Anita
112	101	BDA	2026-02-24	Rahul

(2 rows)

```
cqlsh:library> SELECT * FROM Library_Counter;
```

stud_id	book_name	counter_value
112	BDA	1

(1 rows)

```
cqlsh:library> UPDATE Library_Counter
... SET Counter_value = Counter_value + 1
... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library> SELECT Counter_value
... FROM Library_Counter
... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
```

counter_value
2

(1 rows)

```
cqlsh:library> COPY Library.Library_Info TO 'library_info.csv' WITH HEADER = TRUE;
Using 16 child processes
```

Starting copy of library.library_info with columns [stud_id, book_id, book_name, date_of_issue, stud_name].

Processed: 2 rows; Rate: 14 rows/s; Avg. rate: 14 rows/s
2 rows exported to 1 files in 0.176 seconds.

```
cqlsh:library> COPY Library.Library_Info (Stud_Id, Book_Id, Stud_Name, Book_Name,
Date_of_issue) FROM 'library_info.csv' WITH HEADER = TRUE;
Using 16 child processes
```

Starting copy of library.library_info with columns [stud_id, book_id, stud_name, book_name, date_of_issue].

Processed: 2 rows; Rate: 3 rows/s; Avg. rate: 5 rows/s
2 rows imported from 1 files in 0.407 seconds (0 skipped).

```
cqlsh:library> COPY Library.Library_Info (Stud_Id, Book_Id, Stud_Name, Book_Name,
Date_of_issue) FROM 'library_info.csv' WITH HEADER = TRUE;
Using 16 child processes
```

Starting copy of library.library_info with columns [stud_id, book_id, stud_name, book_name,

date_of_issue].

Processed: 2 rows; Rate: 3 rows/s; Avg. rate: 5 rows/s

2 rows imported from 1 files in 0.407 seconds (0 skipped).

```
cqlsh> CREATE KEYSPACE Library
... WITH replication = {
...   'class': 'SimpleStrategy',
...   'replication_factor': 1
... };
cqlsh> USE Library;
cqlsh:library> CREATE TABLE Library_Info (
...   Stud_Id INT,
...   Book_Id INT,
...   Stud_Name TEXT,
...   Book_Name TEXT,
...   Date_of_issue DATE,
...   PRIMARY KEY (Stud_Id, Book_Id)
... );
cqlsh:library> CREATE TABLE Library_Counter (
...   Stud_Id INT,
...   Book_Name TEXT,
...   Counter_value COUNTER,
...   PRIMARY KEY (Stud_Id, Book_Name)
... );
cqlsh:library> BEGIN BATCH
...
... INSERT INTO Library_Info (Stud_Id, Book_Id, Stud_Name, Book_Name, Date_of_issue)
... VALUES (112, 101, 'Rahul', 'BDA', '2026-02-24');
...
... INSERT INTO Library_Info (Stud_Id, Book_Id, Stud_Name, Book_Name, Date_of_issue)
... VALUES (113, 102, 'Anita', 'AI', '2026-02-24');
...
... APPLY BATCH;
cqlsh:library> UPDATE Library_Counter
... SET Counter_value = Counter_value + 1
... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library> SELECT * FROM Library_Info;

stud_id | book_id | book_name | date_of_issue | stud_name
-----+-----+-----+-----+-----
113 | 102 | AI | 2026-02-24 | Anita
112 | 101 | BDA | 2026-02-24 | Rahul
(2 rows)
cqlsh:library> SELECT * FROM Library_Counter;

stud_id | book_name | counter_value
-----+-----+-----
112 | BDA | 1
(1 rows)
cqlsh:library> UPDATE Library_Counter
... SET Counter_value = Counter_value + 1
... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library> SELECT Counter_value
... FROM Library_Counter
... WHERE Stud_Id = 112 AND Book_Name = 'BDA';

counter_value
-----
2
(1 rows)
cqlsh:library> COPY Library.Library_Info TO 'library_info.csv' WITH HEADER = TRUE;
Using 16 child processes

Starting copy of library.library_info with columns [stud_id, book_id, book_name, date_of_issue, stud_name].
Processed: 2 rows; Rate: 14 rows/s; Avg. rate: 14 rows/s
2 rows exported to 1 files in 0.176 seconds.
cqlsh:library> COPY Library.Library_Info (Stud_Id, Book_Id, Stud_Name, Book_Name, Date_of_issue)
... FROM 'library_info.csv' WITH HEADER = TRUE;
Using 16 child processes

Starting copy of library.library_info with columns [stud_id, book_id, stud_name, book_name, date_of_issue].
Failed to import 1 rows: ParseError - Failed to parse Anita : time data 'Anita' does not match format '%Y-%m-%d', given up without retries
Failed to import 1 rows: ParseError - Failed to parse Rahul : time data 'Rahul' does not match format '%Y-%m-%d', given up without retries
Failed to process 2 rows; failed rows written to 'import_library_info.err'
Processed: 2 rows; Rate: 3 rows/s; Avg. rate: 5 rows/s
0 rows imported from 1 files in 0.406 seconds (0 skipped).
cqlsh:library> COPY Library.Library_Info (Stud_Id, Book_Id, Stud_Name, Book_Name, Date_of_issue) FROM 'library_info.csv' WITH HEADER = TRUE;
Using 16 child processes

Starting copy of library.library_info with columns [stud_id, book_id, stud_name, book_name, date_of_issue].
Processed: 2 rows; Rate: 3 rows/s; Avg. rate: 5 rows/s
2 rows imported from 1 files in 0.407 seconds (0 skipped).
cqlsh:library> select * from library_info;
InvalidRequest: Error from server: code=2200 [Invalid query] message="table library_info does not exist"
cqlsh:library> select * from library_Info;

stud_id | book_id | book_name | date_of_issue | stud_name
-----+-----+-----+-----+-----
113 | 102 | AI | 2026-02-24 | Anita
112 | 101 | BDA | 2026-02-24 | Rahul
```


LABORATORY PROGRAM – 4

Execution of HDFS Commands for interaction with Hadoop Environment. (Minimum 10 commands to be executed)

COMMAND WITH OUTPUT

```
hadoop@bmscsece-HP-Elite-Tower-600-G9-Desktop-PC: $ start-all.sh # Start all Hadoop services
jps # Verify all daemon processes are running.
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
localhost: namenode is running as process 9879. Stop it first and ensure /tmp/hadoop-hadoop-namenode.pid file is empty before retry.
Starting datanodes
localhost: datanode is running as process 10065. Stop it first and ensure /tmp/hadoop-hadoop-datanode.pid file is empty before retry.
Starting secondary namenodes [bmscsece-HP-Elite-Tower-600-G9-Desktop-PC]
bmscsece-HP-Elite-Tower-600-G9-Desktop-PC: secondarynamenode is running as process 10364. Stop it first and ensure /tmp/hadoop-hadoop-secondarynamenode.pid file is empty before retry.
Starting resourceanagers
resourceananager is running as process 10650. Stop it first and ensure /tmp/hadoop-hadoop-resourceananager.pid file is empty before retry.
Starting nodemanagers
localhost: nodemanager is running as process 10825. Stop it first and ensure /tmp/hadoop-hadoop-nodemanager.pid file is empty before retry.
10065 DataNode
16098 Jps
9879 NameNode
10825 NodeManager
10650 ResourceManager
10364 SecondaryNameNode
hadoop@bmscsece-HP-Elite-Tower-600-G9-Desktop-PC: $ hdfs dfs -mkdir /bda_hadoop
hadoop@bmscsece-HP-Elite-Tower-600-G9-Desktop-PC: $ hadoop fs -ls /
Found 23 items
drwxr-xr-x - hadoop supergroup 0 2025-04-16 11:42 /FFF
drwxr-xr-x - hadoop supergroup 0 2023-06-26 14:24 /Pooja134
drwxr-xr-x - hadoop supergroup 0 2023-06-26 15:11 /PoojaBda
drwxr-xr-x - hadoop supergroup 0 2023-05-15 14:24 /sample
drwxr-xr-x - hadoop supergroup 0 2026-03-31 16:15 /bda_hadoop
drwxr-xr-x - hadoop supergroup 0 2023-06-26 15:18 /bda_hadoop
drwxr-xr-x - hadoop supergroup 0 2023-05-22 14:48 /dirnew
drwxr-xr-x - hadoop supergroup 0 2026-03-11 16:24 /inputbda
drwxr-xr-x - hadoop supergroup 0 2023-05-15 15:21 /labB4PB
drwxr-xr-x - hadoop supergroup 0 2025-05-27 11:50 /labcie1
drwxr-xr-x - hadoop supergroup 0 2025-04-16 11:47 /lil
drwxr-xr-x - hadoop supergroup 0 2026-03-31 15:48 /ncdc
drwxr-xr-x - hadoop supergroup 0 2025-05-07 12:06 /output
drwxr-xr-x - hadoop supergroup 0 2026-03-11 16:30 /outputbda
drwxr-xr-x - hadoop supergroup 0 2023-06-26 14:22 /pooja134
-rw-r--r-- 1 hadoop supergroup 52 2023-06-26 15:24 /poojaiyer134
drwxr-xr-x - hadoop supergroup 0 2025-05-07 12:53 /qwe
drwxr-xr-x - hadoop supergroup 0 2025-05-07 11:50 /qwerty
drwxr-xr-x - hadoop supergroup 0 2023-05-17 10:29 /rfs
drwxr-xr-x - hadoop supergroup 0 2023-05-22 14:51 /user
drwxr-xr-x - hadoop supergroup 0 2023-05-22 14:34 /wc_107
drwxr-xr-x - hadoop supergroup 0 2026-03-13 15:45 /weather
drwxr-xr-x - hadoop supergroup 0 2026-03-13 15:55 /weather1
hadoop@bmscsece-HP-Elite-Tower-600-G9-Desktop-PC: $ hdfs dfs -put /home/hduser/Desktop/bda_local.txt /bda_hadoop/file.txt
hdfs dfs -cat /bda_hadoop/file.txt
put: '/home/hduser/Desktop/bda_local.txt': No such file or directory
cat: '/bda_hadoop/file.txt': No such file or directory
hadoop@bmscsece-HP-Elite-Tower-600-G9-Desktop-PC: $ hdfs dfs -mkdir /bda_hadoop
mkdir: '/bda_hadoop': File exists
hadoop@bmscsece-HP-Elite-Tower-600-G9-Desktop-PC: $ echo "Hello Hadoop BDA Lab" > /home/hadoop/Desktop/bda_local.txt
```

```
hadoop@bmscsece-HP-Elite-Tower-600-G9-Desktop-PC: $ hdfs dfs -put /home/hduser/Desktop/bda_local.txt /bda_hadoop/file.txt
hdfs dfs -cat /bda_hadoop/file.txt
put: '/home/hduser/Desktop/bda_local.txt': No such file or directory
cat: '/bda_hadoop/file.txt': No such file or directory
hadoop@bmscsece-HP-Elite-Tower-600-G9-Desktop-PC: $ hdfs dfs -mkdir /bda_hadoop
mkdir: '/bda_hadoop': File exists
hadoop@bmscsece-HP-Elite-Tower-600-G9-Desktop-PC: $ echo "Hello Hadoop BDA Lab" > /home/hadoop/Desktop/bda_local.txt
hadoop@bmscsece-HP-Elite-Tower-600-G9-Desktop-PC: $ cat /home/hadoop/Desktop/bda_local.txt
Hello Hadoop BDA Lab
hadoop@bmscsece-HP-Elite-Tower-600-G9-Desktop-PC: $ hdfs dfs -put /home/hduser/Desktop/bda_local.txt /bda_hadoop/file.txt
hdfs dfs -cat /bda_hadoop/file.txt
put: '/home/hduser/Desktop/bda_local.txt': No such file or directory
cat: '/bda_hadoop/file.txt': No such file or directory
hadoop@bmscsece-HP-Elite-Tower-600-G9-Desktop-PC: $ # put
hdfs dfs -put /home/hadoop/Desktop/bda_local.txt /bda_hadoop/file.txt
hadoop@bmscsece-HP-Elite-Tower-600-G9-Desktop-PC: $ # cat
hdfs dfs -cat /bda_hadoop/file.txt
Hello Hadoop BDA Lab
hadoop@bmscsece-HP-Elite-Tower-600-G9-Desktop-PC: $ hdfs dfs -copyFromLocal /home/hadoop/Desktop/bda_local.txt /bda_hadoop/file_cp_local.txt
hdfs dfs -cat /bda_hadoop/file_cp_local.txt
Hello Hadoop BDA Lab
hadoop@bmscsece-HP-Elite-Tower-600-G9-Desktop-PC: $ # get
hdfs dfs -get /bda_hadoop/file.txt /home/hadoop/Downloads/downloaded_file.txt
hadoop@bmscsece-HP-Elite-Tower-600-G9-Desktop-PC: $
# getmerge
hdfs dfs -getmerge /bda_hadoop/file.txt /bda_hadoop/file_cp_local.txt /home/hadoop/Downloads/merged.txt
hadoop@bmscsece-HP-Elite-Tower-600-G9-Desktop-PC: $ # getfacl
hadoop fs -getfacl /bda_hadoop/
# file: /bda_hadoop/
# owner: hadoop
# group: supergroup
user::rwx
group::r-x
other::r-x
hadoop@bmscsece-HP-Elite-Tower-600-G9-Desktop-PC: $ # copyToLocal
hdfs dfs -copyToLocal /bda_hadoop/file.txt /home/hadoop/Desktop
hadoop@bmscsece-HP-Elite-Tower-600-G9-Desktop-PC: $ # mv
hadoop fs -mv /bda_hadoop /abc
hadoop fs -ls /abc
Found 2 items
-rw-r--r-- 1 hadoop supergroup 21 2026-03-31 16:18 /abc/file.txt
-rw-r--r-- 1 hadoop supergroup 21 2026-03-31 16:18 /abc/file_cp_local.txt
hadoop@bmscsece-HP-Elite-Tower-600-G9-Desktop-PC: $ # cp
hadoop fs -cp /abc /hadoop_lab
hadoop fs -ls /hadoop_lab
Found 2 items
-rw-r--r-- 1 hadoop supergroup 21 2026-03-31 16:19 /hadoop_lab/file.txt
-rw-r--r-- 1 hadoop supergroup 21 2026-03-31 16:19 /hadoop_lab/file_cp_local.txt
hadoop@bmscsece-HP-Elite-Tower-600-G9-Desktop-PC: $
```


Code:

```
hadoop@bmscscse-HP-Elite-Tower-600-G9-Desktop-PC:~$ start-all.sh    # Start all Hadoop services
jps                        # Verify all daemon processes are running
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
localhost: namenode is running as process 9879. Stop it first and ensure /tmp/hadoop-hadoop-namenode.pid file is
empty before retry.
Starting datanodes
localhost: datanode is running as process 10065. Stop it first and ensure /tmp/hadoop-hadoop-datanode.pid file is empty
before retry.
Starting secondary namenodes [bmscscse-HP-Elite-Tower-600-G9-Desktop-PC]
bmscscse-HP-Elite-Tower-600-G9-Desktop-PC: secondarynamenode is running as process 10364. Stop it first and
ensure /tmp/hadoop-hadoop-secondarynamenode.pid file is empty before retry.
Starting resourcemanager
resourcemanager is running as process 10650. Stop it first and ensure /tmp/hadoop-hadoop-resourcemanager.pid file is
empty before retry.
Starting nodemanagers
localhost: nodemanager is running as process 10825. Stop it first and ensure /tmp/hadoop-hadoop-nodemanager.pid file
is empty before retry.
10065 DataNode
16098 Jps
9879 NameNode
10825 NodeManager
10650 ResourceManager
10364 SecondaryNameNode
hadoop@bmscscse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hdfs dfs -mkdir /bda_hadoop
hadoop@bmscscse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hadoop fs -ls /
Found 23 items
drwxr-xr-x - hadoop supergroup      0 2025-04-16 11:42 /FFF
drwxr-xr-x - hadoop supergroup      0 2023-06-26 14:24 /Pooja134
drwxr-xr-x - hadoop supergroup      0 2023-06-26 15:11 /PoojaBda
drwxr-xr-x - hadoop supergroup      0 2023-05-15 14:24 /Sample
drwxr-xr-x - hadoop supergroup      0 2026-03-31 16:15 /bda_hadoop
drwxr-xr-x - hadoop supergroup      0 2023-06-26 15:18 /bdahadoop
drwxr-xr-x - hadoop supergroup      0 2023-05-22 14:48 /dirnew
drwxr-xr-x - hadoop supergroup      0 2026-03-11 16:24 /inputbda
drwxr-xr-x - hadoop supergroup      0 2023-05-15 15:21 /labB4PB
drwxr-xr-x - hadoop supergroup      0 2025-05-27 11:50 /labciel
drwxr-xr-x - hadoop supergroup      0 2025-04-16 11:47 /lll
drwxr-xr-x - hadoop supergroup      0 2026-03-31 15:48 /ncdc
drwxr-xr-x - hadoop supergroup      0 2025-05-07 12:06 /output
drwxr-xr-x - hadoop supergroup      0 2026-03-11 16:30 /outputbda
drwxr-xr-x - hadoop supergroup      0 2023-06-26 14:22 /pooja134
-rw-r--r-- 1 hadoop supergroup    52 2023-06-26 15:24 /poojaiyer134
drwxr-xr-x - hadoop supergroup      0 2025-05-07 12:53 /qwe
drwxr-xr-x - hadoop supergroup      0 2025-05-07 11:56 /qwerty
drwxr-xr-x - hadoop supergroup      0 2023-05-17 10:29 /rgs
drwxr-xr-x - hadoop supergroup      0 2023-05-22 14:51 /user
drwxr-xr-x - hadoop supergroup      0 2023-05-22 14:34 /wc_107
drwxr-xr-x - hadoop supergroup      0 2026-03-13 15:45 /weather
drwxr-xr-x - hadoop supergroup      0 2026-03-13 15:55 /weather1
hadoop@bmscscse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hdfs dfs -put /home/hduser/Desktop/bda_local.txt
/bda_hadoop/file.txt
```

```

hdfs dfs -cat /bda_hadoop/file.txt
put: '/home/hduser/Desktop/bda_local.txt': No such file or directory
cat: '/bda_hadoop/file.txt': No such file or directory
hadoop@bmscece-HP-Elite-Tower-600-G9-Desktop-PC:~$ hdfs dfs -mkdir /bda_hadoop
mkdir: '/bda_hadoop': File exists
hadoop@bmscece-HP-Elite-Tower-600-G9-Desktop-PC:~$ echo "Hello Hadoop BDA Lab" >
/home/hadoop/Desktop/bda_local.txt
hadoop@bmscece-HP-Elite-Tower-600-G9-Desktop-PC:~$ cat /home/hadoop/Desktop/bda_local.txt
Hello Hadoop BDA Lab
hadoop@bmscece-HP-Elite-Tower-600-G9-Desktop-PC:~$ hdfs dfs -put /home/hduser/Desktop/bda_local.txt
/bda_hadoop/file.txt
hdfs dfs -cat /bda_hadoop/file.txt
put: '/home/hduser/Desktop/bda_local.txt': No such file or directory
cat: '/bda_hadoop/file.txt': No such file or directory
hadoop@bmscece-HP-Elite-Tower-600-G9-Desktop-PC:~$ # put
hdfs dfs -put /home/hadoop/Desktop/bda_local.txt /bda_hadoop/file.txt
hadoop@bmscece-HP-Elite-Tower-600-G9-Desktop-PC:~$ # cat
hdfs dfs -cat /bda_hadoop/file.txt
Hello Hadoop BDA Lab
hadoop@bmscece-HP-Elite-Tower-600-G9-Desktop-PC:~$ hdfs dfs -copyFromLocal
/home/hadoop/Desktop/bda_local.txt /bda_hadoop/file_cp_local.txt
hdfs dfs -cat /bda_hadoop/file_cp_local.txt
Hello Hadoop BDA Lab
hadoop@bmscece-HP-Elite-Tower-600-G9-Desktop-PC:~$ # get
hdfs dfs -get /bda_hadoop/file.txt /home/hadoop/Downloads/downloaded_file.txt
hadoop@bmscece-HP-Elite-Tower-600-G9-Desktop-PC:~$
# getmerge
hdfs dfs -getmerge /bda_hadoop/file.txt /bda_hadoop/file_cp_local.txt /home/hadoop/Downloads/merged.txt
hadoop@bmscece-HP-Elite-Tower-600-G9-Desktop-PC:~$ # getfacl
hadoop fs -getfacl /bda_hadoop/
# file: /bda_hadoop
# owner: hadoop
# group: supergroup
user::rwx
group::r-x
other::r-x

hadoop@bmscece-HP-Elite-Tower-600-G9-Desktop-PC:~$ # copyToLocal
hdfs dfs -copyToLocal /bda_hadoop/file.txt /home/hadoop/Desktop
hadoop@bmscece-HP-Elite-Tower-600-G9-Desktop-PC:~$ # mv
hadoop fs -mv /bda_hadoop/abc
hadoop fs -ls /abc
Found 2 items
-rw-r--r-- 1 hadoop supergroup      21 2026-03-31 16:18 /abc/file.txt
-rw-r--r-- 1 hadoop supergroup      21 2026-03-31 16:18 /abc/file_cp_local.txt
hadoop@bmscece-HP-Elite-Tower-600-G9-Desktop-PC:~$ # cp
hadoop fs -cp /abc /hadoop_lab
hadoop fs -ls /hadoop_lab
Found 2 items
-rw-r--r-- 1 hadoop supergroup      21 2026-03-31 16:19 /hadoop_lab/file.txt
-rw-r--r-- 1 hadoop supergroup      21 2026-03-31 16:19 /hadoop_lab/file_cp_local.txt
hadoop@bmscece-HP-Elite-Tower-600-G9-Desktop-PC:~$

```

LABORATORY PROGRAM – 5

Implement Wordcount program on Hadoop framework

CODE, COMMAND WITH OUTPUT

Driver Code

```
// Importing libraries
import java.io.IOException;
import org.apache.hadoop.conf.Configured;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.FileInputFormat;
import org.apache.hadoop.mapred.FileOutputFormat;
import org.apache.hadoop.mapred.JobClient;
import org.apache.hadoop.mapred.JobConf;
import org.apache.hadoop.util.Tool;
import org.apache.hadoop.util.ToolRunner;

public class WCDriver extends Configured implements Tool {

    public int run(String[] args) throws IOException {
        if (args.length < 2) {
            System.out.println("Please give valid inputs");
            return -1;
        }

        JobConf conf = new JobConf(WCDriver.class);
        conf.setJobName("WordCount");

        FileInputFormat.setInputPaths(conf, new Path(args[0]));
        FileOutputFormat.setOutputPath(conf, new Path(args[1]));

        conf.setMapperClass(WCMapper.class);
        conf.setReducerClass(WCReducer.class);

        conf.setMapOutputKeyClass(Text.class);
        conf.setMapOutputValueClass(IntWritable.class);

        conf.setOutputKeyClass(Text.class);
        conf.setOutputValueClass(IntWritable.class);

        JobClient.runJob(conf);
        return 0;
    }

    // Main Method
    public static void main(String[] args) throws Exception {
        int exitCode = ToolRunner.run(new WCDriver(), args);
        System.out.println("Job Exit Code: " + exitCode);
    }
}
```

Mapper Code

```
// Importing libraries
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.Mapper;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reporter;
```

```

public class WCMapper extends MapReduceBase implements Mapper<LongWritable, Text, Text, IntWritable> {

    // Map function
    public void map(LongWritable key, Text value, OutputCollector<Text, IntWritable> output, Reporter reporter)
        throws IOException {
        String line = value.toString();

        // Splitting the line on whitespace
        for (String word : line.split("\\s+")) {
            if (word.length() > 0) {
                output.collect(new Text(word), new IntWritable(1));
            }
        }
    }
}

```

Reducer Code

```

// Importing libraries
import java.io.IOException;
import java.util.Iterator;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reducer;
import org.apache.hadoop.mapred.Reporter;

public class WCReducer extends MapReduceBase implements Reducer<Text, IntWritable, Text, IntWritable> {

    // Reduce function
    public void reduce(Text key, Iterator<IntWritable> values,
        OutputCollector<Text, IntWritable> output,
        Reporter reporter) throws IOException {
        int count = 0;

        // Counting the frequency of each word
        while (values.hasNext()) {
            count += values.next().get();
        }

        output.collect(key, new IntWritable(count));
    }
}

```

```

-rw-r--r-- 1 hadoop supergroup      89 2026-03-11 16:24 /inputbda/sample.txt
hadoop@bmsccese-HP-Elite-Tower-600-G9-Desktop-PC: /wordcount$ hadoop jar ~/Desktop/wordcount.jar WCDriver /inputbda/sample.txt /outputbda
2026-03-11 16:31:13,347 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-03-11 16:31:13,388 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-03-11 16:31:13,388 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-03-11 16:31:13,396 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
Exception in thread "main" org.apache.hadoop.mapred.FileAlreadyExistsException: Output directory hdfs://localhost:9000/outputbda already exists
    at org.apache.hadoop.mapred.FileOutputFormat.checkOutputSpecs(FileOutputFormat.java:131)
    at org.apache.hadoop.mapreduce.JobSubmitter.checkSpecs(JobSubmitter.java:279)
    at org.apache.hadoop.mapreduce.JobSubmitter.submitJobInternal(JobSubmitter.java:143)
    at org.apache.hadoop.mapreduce.Job$11.run(Job.java:1571)
    at org.apache.hadoop.mapreduce.Job$11.run(Job.java:1568)
    at java.base/java.security.AccessController.doPrivileged(Native Method)
    at java.base/javax.security.auth.Subject.doAs(Subject.java:423)
    at org.apache.hadoop.security.UserGroupInformation.doAs(UserGroupInformation.java:1878)
    at org.apache.hadoop.mapreduce.Job.submit(Job.java:1568)
    at org.apache.hadoop.mapred.JobClient$1.run(JobClient.java:576)
    at org.apache.hadoop.mapred.JobClient$1.run(JobClient.java:571)
    at java.base/java.security.AccessController.doPrivileged(Native Method)
    at java.base/javax.security.auth.Subject.doAs(Subject.java:423)
    at org.apache.hadoop.security.UserGroupInformation.doAs(UserGroupInformation.java:1878)
    at org.apache.hadoop.mapred.JobClient.submitJobInternal(JobClient.java:571)
    at org.apache.hadoop.mapred.JobClient.submitJob(JobClient.java:562)
    at org.apache.hadoop.mapred.JobClient.runJob(JobClient.java:873)
    at WCDriver.run(WCDriver.java:28)
    at org.apache.hadoop.util.ToolRunner.run(ToolRunner.java:81)
    at org.apache.hadoop.util.ToolRunner.run(ToolRunner.java:95)
    at WCDriver.main(WCDriver.java:32)
    at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke0(Native Method)
    at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke(NativeMethodAccessorImpl.java:62)
    at java.base/jdk.internal.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorImpl.java:43)
    at java.base/java.lang.reflect.Method.invoke(Method.java:566)
    at org.apache.hadoop.util.RunJar.run(RunJar.java:323)
    at org.apache.hadoop.util.RunJar.main(RunJar.java:236)
hadoop@bmsccese-HP-Elite-Tower-600-G9-Desktop-PC: /wordcount$ hadoop fs -ls /outputbda
hadoop fs -cat /outputbda/part-00000
Found 2 items
-rw-r--r-- 1 hadoop supergroup      0 2026-03-11 16:30 /outputbda/_SUCCESS
-rw-r--r-- 1 hadoop supergroup    69 2026-03-11 16:30 /outputbda/part-00000
are 1
brother 1
family 1
ht 1
how 5
is 4
job 1
sister 1
you 1
your 4
hadoop@bmsccese-HP-Elite-Tower-600-G9-Desktop-PC: /wordcount$

```

LABORATORY PROGRAM – 6

Implement Weather program on Hadoop framework

Questions:

From the following link extract the weather data

<https://github.com/tomwhite/hadoopbook/tree/master/input/ncdc/all>

- a) Create a MapReduce program to find average temperature for each year from NCDC data set.
- b) find the mean max temperature for every month.

CODE, COMMAND WITH OUTPUT – A

Driver Code

```
package temp;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;

import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class AverageDriver {

    public static void main(String[] args) throws Exception {

        if (args.length != 2) {
            System.err.println("Please enter both input and output parameters.");
            System.exit(-1);
        }

        // Creating a configuration and job instance
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Average Calculation");

        job.setJarByClass(AverageDriver.class);

        // Input and output paths
        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));

        // Setting mapper and reducer classes
        job.setMapperClass(AverageMapper.class);
        job.setReducerClass(AverageReducer.class);

        // Output key and value types
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);

        // Submitting the job and waiting for it to complete
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

Mapper Code

```
package temp;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class AverageMapper extends Mapper<LongWritable, Text, Text, IntWritable> {

    public static final int MISSING = 9999;

    @Override
    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {

        String line = value.toString();

        // Extract year from fixed position
        String year = line.substring(15, 19);
        int temperature;

        // Determine if there's a '+' sign
        if (line.charAt(87) == '+') {
            temperature = Integer.parseInt(line.substring(88, 92));
        } else {
            temperature = Integer.parseInt(line.substring(87, 92));
        }

        // Quality check character
        String quality = line.substring(92, 93);

        // Only emit if data is valid
        if (temperature != MISSING && quality.matches("[01459]")) {
            context.write(new Text(year), new IntWritable(temperature));
        }
    }
}
```

Reducer Code

```
package temp;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class AverageReducer extends Reducer<Text, IntWritable, Text, IntWritable> {

    @Override
    public void reduce(Text key, Iterable<IntWritable> values,
        Context context) throws IOException, InterruptedException {

        int sumTemp = 0;
        int count = 0;

        for (IntWritable value : values) {
            sumTemp += value.get();
            count++;
        }

        if (count > 0) {
            int average = sumTemp / count;
            context.write(key, new IntWritable(average));
        }
    }
}
```

```
}  
}  
}
```

```
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/ncdc$ hdfs dfs -cat /ncdc/output-avg/part-r-00000  
1901      4.67  
1902      2.17  
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/ncdc$ hdfs dfs -cat /ncdc/output-maxmonth/part-r-00000  
1901-01 0.77  
1901-02 -2.4  
1901-03 0.47  
1901-04 7.53  
1901-05 15.13  
1901-06 21.37  
1901-07 26.69  
1901-08 22.71  
1901-09 16.58  
1901-10 11.51  
1901-11 3.26  
1901-12 -0.5  
1902-01 0.54  
1902-02 -0.4  
1902-03 0.07  
1902-04 4.28  
1902-05 12.35  
1902-06 17.49  
1902-07 18.29  
1902-08 17.1  
1902-09 12.39  
1902-10 6.98  
1902-11 3.57  
1902-12 0.22  
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/ncdc$
```


CODE, COMMAND WITH OUTPUT – B

Driver Code

```
package meanmax;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;

import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class MeanMaxDriver {

    public static void main(String[] args) throws Exception {

        if (args.length != 2) {
            System.err.println("Please enter both input and output parameters.");
            System.exit(-1);
        }

        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Mean and Max Temperature");

        job.setJarByClass(MeanMaxDriver.class);

        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));

        job.setMapperClass(MeanMaxMapper.class);
        job.setReducerClass(MeanMaxReducer.class);

        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);

        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

Mapper Code

```
package meanmax;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Mapper;

public class MeanMaxMapper extends Mapper<LongWritable, Text, Text, IntWritable> {

    public static final int MISSING = 9999;

    @Override
    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {

        String line = value.toString();

        // Extract month from positions 19-20
        String month = line.substring(19, 21);
        int temperature;
```

```

// Extract temperature considering optional '+'
if (line.charAt(87) == '+') {
    temperature = Integer.parseInt(line.substring(88, 92));
} else {
    temperature = Integer.parseInt(line.substring(87, 92));
}

// Quality check
String quality = line.substring(92, 93);

if (temperature != MISSING && quality.matches("[01459]")) {
    context.write(new Text(month), new IntWritable(temperature));
}
}
}

```

Reducer Code

```

package meanmax;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class MeanMaxReducer extends Reducer<Text, IntWritable, Text, Text> {

    @Override
    public void reduce(Text key, Iterable<IntWritable> values,
        Context context) throws IOException, InterruptedException {

        int sumTemp = 0;
        int count = 0;
        int maxTemp = Integer.MIN_VALUE;

        for (IntWritable value : values) {
            int temp = value.get();
            sumTemp += temp;
            count++;

            if (temp > maxTemp) {
                maxTemp = temp;
            }
        }

        if (count > 0) {
            int avgTemp = sumTemp / count;
            String result = "mean=" + avgTemp + " max=" + maxTemp;
            context.write(key, new Text(result));
        }
    }
}

```

```

hadoop@msciece-HP-Elite-Tower-800-G9-Desktop-PC: $ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
localhost: namenode is running as process 5478. Stop it first and ensure /tmp/hadoop-hadoop-namenode.pid file is empty before retry.
Starting datanodes
localhost: datanode is running as process 5644. Stop it first and ensure /tmp/hadoop-hadoop-datanode.pid file is empty before retry.
Starting secondary namenodes [msciece-HP-Elite-Tower-800-G9-Desktop-PC]
msciece-HP-Elite-Tower-800-G9-Desktop-PC: secondarynamenode is running as process 5931. Stop it first and ensure /tmp/hadoop-hadoop-secondarynamenode.pid file is empty before retry.
Starting resourcemanager
resourcemanager is running as process 6214. Stop it first and ensure /tmp/hadoop-hadoop-resourcemanager.pid file is empty before retry.
Starting nodemanager
localhost: nodemanager is running as process 6375. Stop it first and ensure /tmp/hadoop-hadoop-nodemanager.pid file is empty before retry.
hadoop@msciece-HP-Elite-Tower-800-G9-Desktop-PC: $ hdfs dfs -copyFromLocal /home/hadoop/Desktop/1901 /rgs/temp
copyFromLocal: /rgs/temp: File exists
hadoop@msciece-HP-Elite-Tower-800-G9-Desktop-PC: $ hdfs dfs -mkdir -p /rgs
hadoop@msciece-HP-Elite-Tower-800-G9-Desktop-PC: $ hdfs dfs -copyFromLocal /home/hadoop/Desktop/1901 /rgs/1903
hadoop@msciece-HP-Elite-Tower-800-G9-Desktop-PC: $ hadoop jar /home/hduser/Downloads/Map_Max_Temp.jar MMRDriver /rgs/avtemp.txt /out8
JAR does not exist or is not a normal file: /home/hduser/Downloads/Map_Max_Temp.jar
hadoop@msciece-HP-Elite-Tower-800-G9-Desktop-PC: $ hadoop jar /home/hduser/Downloads/Min_Max_Temp.jar MMRDriver /rgs/avtemp.txt /out8
JAR does not exist or is not a normal file: /home/hduser/Downloads/Min_Max_Temp.jar
hadoop@msciece-HP-Elite-Tower-800-G9-Desktop-PC: $ hadoop jar /home/hadoop/Desktop/Min_Max_Temp.jar MMRDriver /rgs/avtemp.txt /out8
2025-05-06 15:23:05,430 INFO Impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2025-05-06 15:23:05,471 INFO Impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2025-05-06 15:23:05,471 INFO Impl.MetricsSystemImpl: JobTracker metrics system started
2025-05-06 15:23:05,531 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2025-05-06 15:23:05,575 INFO mapreduce.JobSubmitter: Cleaning up the staging area file:/tmp/hadoop/mapred/staging/hadoop1762085270/.staging/job_local1762085270_0001
Exception in thread "main" org.apache.hadoop.mapreduce.lib.input.InvalidInputException: Input path does not exist: hdfs://localhost:9000/rgs/avtemp.txt
    at org.apache.hadoop.mapreduce.lib.input.FileInputFormat.singleThreadedListStatus(FileInputFormat.java:346)
    at org.apache.hadoop.mapreduce.lib.input.FileInputFormat.listStatus(FileInputFormat.java:279)
    at org.apache.hadoop.mapreduce.lib.input.FileInputFormat.getSplit(FileInputFormat.java:404)
    at org.apache.hadoop.mapreduce.JobSubmitter.writeNewSplits(JobSubmitter.java:310)
    at org.apache.hadoop.mapreduce.JobSubmitter.writeSplits(JobSubmitter.java:327)
    at org.apache.hadoop.mapreduce.JobSubmitter.submitJobInternal(JobSubmitter.java:280)
    at org.apache.hadoop.mapreduce.Job$1.run(Job.java:1678)
    at org.apache.hadoop.mapreduce.Job$1.run(Job.java:1675)
    at java.base/java.security.AccessController.doPrivileged(Native Method)
    at java.base/javax.security.auth.Subject.doAs(Subject.java:423)
    at org.apache.hadoop.security.UserGroupInformation.doAs(UserGroupInformation.java:1899)
    at org.apache.hadoop.mapreduce.Job.submit(Job.java:1675)
    at org.apache.hadoop.mapreduce.Job.waitForCompletion(Job.java:1696)
    at MMRDriver.main(MMRDriver.java:40)
    at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke0(Native Method)
    at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke(NativeMethodAccessorImpl.java:62)
    at java.base/jdk.internal.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorImpl.java:43)
    at java.base/java.lang.reflect.Method.invoke(Method.java:566)
    at org.apache.hadoop.util.RunJar.run(RunJar.java:328)
    at org.apache.hadoop.util.RunJar.main(RunJar.java:241)
Caused by: java.io.IOException: Input path does not exist: hdfs://localhost:9000/rgs/avtemp.txt
    at org.apache.hadoop.mapreduce.lib.input.FileInputFormat.singleThreadedListStatus(FileInputFormat.java:313)
    ... 19 more
hadoop@msciece-HP-Elite-Tower-800-G9-Desktop-PC: $ hdfs dfs -cat /out8/*
cat: /out8/*: No such file or directory
hadoop@msciece-HP-Elite-Tower-800-G9-Desktop-PC: $ ^C

```

```

Caused by: java.io.IOException: Input path does not exist: hdfs://localhost:9000/rgs/avtemp.txt
    at org.apache.hadoop.mapreduce.lib.input.FileInputFormat.singleThreadedListStatus(FileInputFormat.java:313)
    ... 19 more
hadoop@msciece-HP-Elite-Tower-800-G9-Desktop-PC: $ hadoop jar /home/hadoop/Desktop/Min_Max_Temp.jar MMRDriver /rgs/1903 /out8
2025-05-06 15:26:34,876 INFO Impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2025-05-06 15:26:34,916 INFO Impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2025-05-06 15:26:34,916 INFO Impl.MetricsSystemImpl: JobTracker metrics system started
2025-05-06 15:26:34,976 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2025-05-06 15:26:35,029 INFO Input.FileInputFormat: Total input files to process : 1
2025-05-06 15:26:35,081 INFO mapreduce.JobSubmitter: number of splits:1
2025-05-06 15:26:35,148 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local1063792118_0001
2025-05-06 15:26:35,148 INFO mapreduce.JobSubmitter: Executing with tokens: []
2025-05-06 15:26:35,215 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2025-05-06 15:26:35,216 INFO mapreduce.Job: Running Job: job_local1063792118_0001
2025-05-06 15:26:35,216 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2025-05-06 15:26:35,220 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2025-05-06 15:26:35,221 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2025-05-06 15:26:35,221 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2025-05-06 15:26:35,221 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter
2025-05-06 15:26:35,267 INFO mapred.LocalJobRunner: Waiting for map tasks
2025-05-06 15:26:35,268 INFO mapred.LocalJobRunner: Starting task: attempt_local1063792118_0001_m_000000_0
2025-05-06 15:26:35,278 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2025-05-06 15:26:35,278 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2025-05-06 15:26:35,279 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2025-05-06 15:26:35,286 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2025-05-06 15:26:35,287 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/rgs/1903:0+888190
2025-05-06 15:26:35,322 INFO mapred.MapTask: (EQUATOR) 0 kvl 26214396(104857584)
2025-05-06 15:26:35,322 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2025-05-06 15:26:35,322 INFO mapred.MapTask: soft limit at 83886080
2025-05-06 15:26:35,322 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2025-05-06 15:26:35,322 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2025-05-06 15:26:35,325 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2025-05-06 15:26:35,440 INFO mapred.LocalJobRunner:
2025-05-06 15:26:35,441 INFO mapred.MapTask: Starting flush of map output
2025-05-06 15:26:35,441 INFO mapred.MapTask: Spilling map output
2025-05-06 15:26:35,441 INFO mapred.MapTask: bufstart = 0; bufend = 45948; bufvoid = 104857600
2025-05-06 15:26:35,441 INFO mapred.MapTask: kvstart = 26214396(104857584); kvoid = 26188144(104752576); length = 26253/6553600
2025-05-06 15:26:35,451 INFO mapred.MapTask: Finished spill 0
2025-05-06 15:26:35,456 INFO mapred.Task: Task:attempt_local1063792118_0001_m_000000_0 is done. And is in the process of committing
2025-05-06 15:26:35,458 INFO mapred.LocalJobRunner: map
2025-05-06 15:26:35,458 INFO mapred.Task: Task: 'attempt_local1063792118_0001_m_000000_0' is done.
2025-05-06 15:26:35,461 INFO mapred.Task: Final Counters for attempt_local1063792118_0001_m_000000_0: Counters: 23
File System Counters
  FILE: Number of bytes read=4359
  FILE: Number of bytes written=703803
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=888190
  HDFS: Number of bytes written=0
  HDFS: Number of read operations=5
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=1
  HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
  Map input records=6553

```

```

2025-05-06 15:26:36,233 INFO Mapreduce.Job: Counters: 36
  File System Counters
    FILE: Number of bytes read=126914
    FILE: Number of bytes written=1466688
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=1776380
    HDFS: Number of bytes written=74
    HDFS: Number of read operations=15
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=4
    HDFS: Number of bytes read erasure-coded=0
  Map-Reduce Framework
    Map input records=6565
    Map output records=6564
    Map output bytes=45948
    Map output materialized bytes=59082
    Input split bytes=95
    Combine input records=0
    Combine output records=0
    Reduce input groups=12
    Reduce shuffle bytes=59082
    Reduce input records=6564
    Reduce output records=12
    Spilled Records=13128
    Shuffled Maps =1
    Failed Shuffles=0
    Merged Map outputs=1
    GC time elapsed (ms)=0
    Total committed heap usage (bytes)=1052770304
  Shuffle Errors
    BAD_ID=0
    CONNECTION=0
    IO_ERROR=0
    WRONG_LENGTH=0
    WRONG_MAP=0
    WRONG_REDUCE=0
  File Input Format Counters
    Bytes Read=888190
  File Output Format Counters
    Bytes Written=74

```

```

hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -cat /out8/*
01      4
02      0
03      7
04     44
05    100
06    168
07    219
08    198
09    141
10    100
11     19
12      3

```

LABORATORY PROGRAM – 7

For a given Text file, Create a Map Reduce program to sort the content in an alphabetic order listing only top 10 maximum occurrences of words.

CODE, COMMAND WITH OUTPUT

Driver Code (TopNDriver.java)

```
package samples.topn;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class TopNDriver {

    public static void main(String[] args) throws Exception {
        if (args.length != 3) {
            System.err.println("Usage: TopNDriver <in> <temp-out> <final-out>");
            System.exit(2);
        }

        Configuration conf = new Configuration();

        // === Job 1: Word Count ===
        Job wcJob = Job.getInstance(conf, "word count");
        wcJob.setJarByClass(TopNDriver.class);
        wcJob.setMapperClass(WordCountMapper.class);
        wcJob.setCombinerClass(WordCountReducer.class);
        wcJob.setReducerClass(WordCountReducer.class);
        wcJob.setOutputKeyClass(Text.class);
        wcJob.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(wcJob, new Path(args[0]));
        Path tempDir = new Path(args[1]);
        FileOutputFormat.setOutputPath(wcJob, tempDir);

        if (!wcJob.waitForCompletion(true)) {
            System.exit(1);
        }

        // === Job 2: Top N ===
        Job topJob = Job.getInstance(conf, "top 10 words");
        topJob.setJarByClass(TopNDriver.class);
        topJob.setMapperClass(TopNMapper.class);
        topJob.setReducerClass(TopNReducer.class);
        topJob.setMapOutputKeyClass(IntWritable.class);
        topJob.setMapOutputValueClass(Text.class);
        topJob.setOutputKeyClass(Text.class);
        topJob.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(topJob, tempDir);
        FileOutputFormat.setOutputPath(topJob, new Path(args[2]));

        System.exit(topJob.waitForCompletion(true) ? 0 : 1);
    }
}
```

Mapper Code (WordCountMapper.java)

```
package samples.topn;

import java.io.IOException;
import java.util.StringTokenizer;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class WordCountMapper
    extends Mapper<Object, Text, Text, IntWritable> {

    private final static IntWritable ONE = new IntWritable(1);
    private Text word = new Text();
    // characters to normalize into spaces
    private String tokens = "[_!$%<>\\^=\\[\\]\\|\\*\\/\\\\\\.,;:\\\\-:()?!\"']";

    @Override
    protected void map(Object key, Text value, Context context)
        throws IOException, InterruptedException {

        // clean & tokenize
        String clean = value.toString()
            .toLowerCase()
            .replaceAll(tokens, " ");
        StringTokenizer itr = new StringTokenizer(clean);
        while (itr.hasMoreTokens()) {
            word.set(itr.nextToken().trim());
            context.write(word, ONE);
        }
    }
}
```

Mapper Code (TopNMapper.java)

```
package samples.topn;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class TopNMapper
    extends Mapper<Object, Text, IntWritable, Text> {

    private IntWritable count = new IntWritable();
    private Text word = new Text();

    @Override
    protected void map(Object key, Text value, Context context)
        throws IOException, InterruptedException {

        // input line: word \t count
        String[] parts = value.toString().split("\\t");
        if (parts.length == 2) {
            word.set(parts[0]);
            count.set(Integer.parseInt(parts[1]));
            // emit count → word, so Hadoop sorts by count
            context.write(count, word);
        }
    }
}
```

Reducer Code (WordCountReducer.java)

```
package samples.topn;

import java.io.IOException;
```

```

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class WordCountReducer
    extends Reducer<Text, IntWritable, Text, IntWritable> {

    @Override
    protected void reduce(Text key, Iterable<IntWritable> values, Context context)
        throws IOException, InterruptedException {

        int sum = 0;
        for (IntWritable val : values) {
            sum += val.get();
        }
        context.write(key, new IntWritable(sum));
    }
}

```

Reducer Code (TopNReducer.java)

```

package samples.topn;

import java.io.IOException;
import java.util.ArrayList;
import java.util.Collections;
import java.util.List;
import java.util.Map;
import java.util.TreeMap;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class TopNReducer
    extends Reducer<IntWritable, Text, Text, IntWritable> {

    // TreeMap with descending order of keys (counts)
    private TreeMap<Integer, List<String>> countMap =
        new TreeMap<>(Collections.reverseOrder());

    @Override
    protected void reduce(IntWritable key, Iterable<Text> values, Context context)
        throws IOException, InterruptedException {

        int cnt = key.get();
        List<String> words = countMap.getOrDefault(cnt, new ArrayList<>());
        for (Text w : values) {
            words.add(w.toString());
        }
        countMap.put(cnt, words);
    }

    @Override
    protected void cleanup(Context context)
        throws IOException, InterruptedException {

        // collect top 10 word→count pairs
        List<WordCount> topList = new ArrayList<>();
        int seen = 0;
        for (Map.Entry<Integer, List<String>> entry : countMap.entrySet()) {
            int cnt = entry.getKey();
            for (String w : entry.getValue()) {
                topList.add(new WordCount(w, cnt));
                seen++;
                if (seen == 10) break;
            }
            if (seen == 10) break;
        }
    }
}

```

```

    }

    // sort these 10 entries alphabetically by word
    Collections.sort(topList, (a, b) -> a.word.compareTo(b.word));

    // emit final top 10 in alphabetical order
    for (WordCount wc : topList) {
        context.write(new Text(wc.word), new IntWritable(wc.count));
    }
}

// helper class
private static class WordCount {
    String word;
    int count;
    WordCount(String w, int c) { word = w; count = c; }
}
}

```

```

bytes written=69
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$ hadoop fs -cat /outputtopn/part-r-00000
how      5
your     4
is       4
brother  1
are      1
hi       1
sister   1
family   1
you      1
job      1
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$

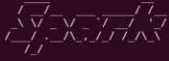
```


LABORATORY PROGRAM – 8

Write a Scala program to print numbers from 1 to 100 using for loop.

CODE, COMMAND WITH OUTPUT

```
object PrintNumbers {  
  def main(args: Array[String]): Unit={  
    for (i<- 1 to 100) {  
      println(i)  
    }  
  }  
}
```

```
bmccesce@bmccesce-HP-Elite-Tower-600-G9-Desktop-PC: $ spark-shell  
25/05/20 11:28:13 WARN Utils: Your hostname, bmccesce-HP-Elite-Tower-600-G9-Desktop-PC resolves to a loopback address: 127.0.1.1; using 10.124.3.80 instead (on interface eno1)  
25/05/20 11:28:13 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address  
WARNING: An illegal reflective access operation has occurred  
WARNING: Illegal reflective access by org.apache.spark.unsafe.Platform (file:/opt/spark/jars/spark-unsafe_2.12-3.0.3.jar) to constructor java.nio.DirectByteBuffer(long,int)  
WARNING: Please consider reporting this to the maintainers of org.apache.spark.unsafe.Platform  
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operations  
WARNING: All illegal access operations will be denied in a future release  
25/05/20 11:28:13 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable  
Using Spark's default log4j profile: org/apache/spark/log4j-defaults.properties  
Setting default log level to "WARN".  
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).  
Spark context Web UI available at http://10.124.3.80:4040  
Spark context available as 'sc' (master = local[*], app id = local-1747720695950).  
Spark session available as 'spark'.  
Welcome to  
 version 3.0.3  
  
Using Scala version 2.12.10 (OpenJDK 64-Bit Server VM, Java 11.0.26)  
Type in expressions to have them evaluated.  
Type :help for more information.  
  
scala> for (i <- 1 to 100) print(i + " ")  
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
```

Using RDD and FlatMap count how many times each word appears in a file and write out a list of words whose count is strictly greater than 4 using Spark.

[illegible]

30 | P a g e

5x2

version 3.0.3

```
scala> val data = sc.textFile("file:///home/bmscscse/input.txt")
data: org.apache.spark.rdd.RDD[String] = file:///home/bmscscse/input.txt MapPartitionsRDD[1] at textFile
at <console>:24
```

```
scala> val words = data.flatMap(line => line.split(" "))
words: org.apache.spark.rdd.RDD[String] = MapPartitionsRDD[2] at flatMap at <console>:25
```

```
scala> val wordPairs = words.map(word => (word, 1))
wordPairs: org.apache.spark.rdd.RDD[(String, Int)] = MapPartitionsRDD[3] at map at <console>:25
```

scala>

```
scala> val wordCounts = wordPairs.reduceByKey(_ + _)
wordCounts: org.apache.spark.rdd.RDD[(String, Int)] = ShuffledRDD[4] at reduceByKey at <console>:25
```

```
scala> val result = wordCounts.filter{ case (word, count) => count > 4 }
result: org.apache.spark.rdd.RDD[(String, Int)] = MapPartitionsRDD[5] at filter at <console>:25
```

```
scala> result.collect().foreach(println)
(hello,7)
```

```
scala> 
```

LABORATORY PROGRAM – 10

Write a simple streaming program in Spark to receive text data streams on a particular port, perform basic text cleaning (like white space removal, stop words removal, lemmatization, etc.), and print the cleaned text on the screen. (Open Ended Question).

CODE, COMMAND WITH OUTPUT

```
# Install NLTK and download required data (run once)
!pip install nltk

import nltk
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')

from pyspark.sql import SparkSession
from pyspark.sql.functions import col, lower, regexp_replace, split, explode, udf
from pyspark.sql.types import ArrayType, StringType
from pyspark.ml.feature import StopWordsRemover
from nltk.stem import WordNetLemmatizer

# Initialize SparkSession
spark = SparkSession.builder.appName("TextProcessing").getOrCreate()

# Define your input lines
lines = [
    "Hello, I hate you.",
    "I hate that I love you.",
    "Don't want to, but I can't put",
    "nobody else above you."
]

# Create DataFrame from lines
df = spark.createDataFrame(lines, "string").toDF("value")

# Step 1: Lowercase and remove punctuation
df_clean = df.select(regexp_replace(lower(col("value")), "[^a-zA-Z\\s]", "").alias("cleaned"))

# Step 2: Tokenize the cleaned text
df_tokens = df_clean.select(split(col("cleaned"), "\\s+").alias("tokens"))

# Step 3: Remove stop words
remover = StopWordsRemover(inputCol="tokens", outputCol="filtered")
df_filtered = remover.transform(df_tokens)

# Step 4: Lemmatization using NLTK WordNetLemmatizer with UDF
lemmatizer = WordNetLemmatizer()

def lemmatize_words(words):
    return [lemmatizer.lemmatize(word) for word in words]

lemmatize_udf = udf(lemmatize_words, ArrayType(StringType()))

df_lemmatized = df_filtered.withColumn("lemmatized", lemmatize_udf(col("filtered")))
```

Step 5: Explode the lemmatized words and show results

```
df_lemmatized.select(explode(col("lemmatized")).alias("word")).show(truncate=False)
```

```
26/04/21 16:18:20 WARN TextSocketSourceProvider: The socket source should not be used for production applications! It does not support recovery.
26/04/21 16:18:20 WARN StreamingQueryManager: Temporary checkpoint location created which is deleted normally when the query didn't fail: /tmp/temporary-aa0bea3b-0517-4341-a46c-aaedc118d895. If it's required to delete it under any circumstances, please set spark.sql.streaming.forceDeleteTempCheckpointLocation to true. Important to know deleting temp checkpoint folder is best effort.
```

```
-----
Batch: 0
```

```
-----+-----+
|original|cleaned|
+-----+-----+
+-----+-----+
```

```
-----
Batch: 1
```

```
-----+-----+-----+
|original|cleaned|
+-----+-----+-----+
|The cats are running quickly over the lazy dogs!!!|cat runn quickly lazy dog|
+-----+-----+-----+
```